

Vishnu Vardhan Addanki Tirumala

Date of birth: 06/03/2002 | **Nationality:** Indian | **Phone number:** (+91) 9866146332 (Mobile) | **Phone number:** (+47) 40581636 (Home) | **Email address:** vishnuvardhanaddanki22@gmail.com | **Website:** https://vishnu-add.github.io/ | **Website:** https://github.com/Vishnu-add/ | **LinkedIn:** A Vishnu Vardhan | **Address:** India (Home)

ABOUT ME

I build AI systems, then try to figure out why they fail. Two years in production ML, building and shipping LLM agents, ML models and RAG pipelines at scale, showed me how often "it works" and "I know why it works" pull apart. I'm now a Master's researcher and University Research Assistant working on interpretability and robustness in vision-language models and Tsetlin Machines, alongside a continued interest in agentic systems.

PUBLICATIONS

2026

Clause Sanitization for Compact and Interpretable Tsetlin Machines

Journal Name: International Symposium on the Tsetlin Machine

EDUCATION AND TRAINING

05/08/2025 – CURRENT Grimstad, Norway

MASTERS IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE Indian Institute of Information Technology, Design and Manufacturing AND University of Agder

Research track in VLMs : Fine-tuning vision-language models with LoRA and evaluating how well they generalize across different data domains.

01/07/2021 – 15/06/2024 Hyderabad, India

BACHELOR OF ENGINEERING IN COMPUTER SCIENCE (AI & ML) Neil Gogte Institute of Technology

Level in EQF EQF level 6

01/07/2017 – 31/05/2021 Hyderabad, India

DIPLOMA IN EMBEDDED SYSTEMS Government Institute of Electronics

Level in EQF EQF level 5

30/04/2017 Hyderabad, India

SECONDARY SCHOOL CERTIFICATE (SSC) St. Anthony's High School

Level in EQF EQF level 4

WORK EXPERIENCE

UNIVERSITY RESEARCH ASSISTANT – UNIVERSITY OF AGDER – 01/01/2026 – Current – AGDER, NORWAY

- Authored two papers submitted to **International Symposium on the Tsetlin Machine** (ISTM) 2026 conference on Tsetlin Machine interpretability. One paper, titled "**Clause Sanitization for Compact and Interpretable Tsetlin Machines"** was accepted for publication.

PRODUCT DEVELOPMENT ENGINEER (MACHINE LEARNING) – PHENOM PEOPLE PVT LTD – 20/06/2024 – 31/07/2025 – HYDERABAD, INDIA

AI - Scheduling & HITL :

1. Led development, deployment, and SRE responsibilities for AI Scheduling microservices.

2. Maintained and optimized a multi-label classification pipeline for multi-agent emails using ML automation and HITL.
3. Improved email classification automation: Candidate emails (30% to 48%), Recruiter/Interviewer emails (0% to 60%) with 95% accuracy.
4. Deployed BERT-based classifiers and built a Auditing system to capture intents, sub-intents and entities, supported by performance dashboards.

AI - Agents Development :

1. Built core logic for a voice-based screening agent handling 1000+ calls monthly.
2. Integrated a 99.9% accurate Voicemail Detection system that eliminated voicemail costs and improved call efficiency.
3. Developed a questionnaire builder agent enabling the product team to generate job-specific questionnaires from job descriptions.
4. Diagnosed and resolved backend integration issues, enhancing agent performance and system stability.
5. Contributed to delivering MVPs for multiple AI agents, accelerating product development.

MACHINE LEARNING INTERN – WADHWANI AI – 10/07/2023 – 19/06/2024 – DELHI, INDIA

Clinical Decision Support System (CDSS):

1. Built clinical NER models to extract medical entities from free-text data.
2. Designed UIs for KB validation; worked with ASR and TTS integrations.

Personal Assistant for Frontline Healthcare Workers:

1. Co-developed a RAG-based chatbot for 1M+ healthcare workers to answer domain-specific queries.
2. Added logic to reduce hallucinations in low-data settings; fine-tuned embedding models and linear adapters.
3. Developed synthetic pipelines and evaluated system performance; integrated vector DBs, ASR, and TTS.
4. Using LLMs, Prompt Engineering, vector stores, ASR and TTS services

JUNIOR MACHINE LEARNING ENGINEER – OMDENA AI – 01/05/2023 – 30/06/2024 – REMOTE

Selected as one of 90 contributors for two global AI projects.

1. **Montreal Chapter – AI-Powered Dubbing App:** Worked on speech-to-text and text-to-speech modules using various AI/NLP libraries to create natural-sounding dubbing.
2. **Toronto Chapter – SmartGuide AI Chatbot:** Built and tested chatbot flows using the Rasa framework, Flask, and FastAPI; configured backend for integration.

MACHINE LEARNING INTERN – CODING RAJA TECHNOLOGIES – 01/11/2023 – 31/12/2023 – REMOTE

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1. Built a Twitter Sentiment Analysis Web App using Flask and NLP models.
2. Fine-tuned a Vision Transformer (ViT) for Indian food image classification with 94% accuracy using Hugging Face transformers.

AI INTERN – AICTE – 01/06/2023 – 31/07/2023 – REMOTE

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1. Developed a fully functional Mental fitness tracker to track and monitor a country's mental disorders percentage using 12 regression and ML models.
2. Performed data cleaning, preprocessing, and feature engineering to improve model performance.

APPRENTICE – ELECTRONICS CORPORATION OF INDIA LTD – 01/06/2020 – 31/05/2021 – HYDERABAD, INDIA

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1. I successfully managed the projects and achieved the scheduled goals.
2. I also contributed to increasing the production of modules and I worked diligently to improve the accuracy of the modules produced.

SKILLS

Machine Learning & Deep Learning

PyTorch | Transformers | Scikit-Learn | Keras | TensorFlow

Languages

Python (computer programming) | SQL | C++ | Java

NLP & GenAI

LLMs | Langchain | Langgraph | SpaCy | NLTK | RAG | Prompt Engineering

Frameworks

FastAPI | Flask | Streamlit | Gradio | Rasa | Chainlit

Databases & Vector Stores

PostgreSQL | MongoDB | Redis | Elasticsearch | Weaviate

DevOps & Infra

Docker | Kubernetes | Jenkins | ArgoCD | Kafka | RabbitMQ

Tools & Others

Git | Linux

LANGUAGE SKILLS

Mother tongue(s): **TELUGU** | **HINDI**

Other language(s): **ENGLISH**

PROJECTS

AI-Assisted Hardware-Software Co-Design Agents (coursework, 2026)

Building custom LLM agents for a hardware and software co-design pipeline, using existing design software as the agents' environment, with human-in-the-loop review built into the workflow.

Sight-Sphere Communicator

Team Lead and Developed an assistive device for visually impaired individuals using ESP32-CAM, ultrasonic sensors, and multimodal LLM on a Raspberry Pi to provide real-time audio navigation instructions.

AI Video Generator

Built a text-to-video generator platform with narration, BGM, scene and character consistency across video using LangChain, FastAPI, TTS models, image and music gen models. Included story creation, video editor, and video and story library.

EduBot: RAG-Based AI University Assistant

Developed an AI chatbot to assist students with university resources through document Q&A and video lecture interaction. Utilized advanced RAG, LangChain, vLLM, and transformers. Led full-cycle development, managed front-end and backend APIs, and explored multi-cloud deployment (Azure, AWS, Render, Vercel).

Face Recognition Attendance System

Team Lead and Developed a face recognition attendance system using Django, leveraging OpenFace for feature extraction and a custom machine learning model for face recognition.

VOLUNTEERING

01/08/2015 – 31/05/2020 India
National Cadet Corps

01/10/2021 – 30/04/2024 India
Head of Dance Club at NGIT

HONOURS AND AWARDS

NFS winner – Neil Gogte Institute of Technology

NFS winner in the weekly coding contest called NFS by NGIT, achieved 3rd position in the 2nd week, 5th position in the 4th week and 4th position in 7th week.

CERTIFICATIONS

IBM Skillsbuild
AI and ML

Mode of learning: Online

EPAM SYSTEMS
Frontend technologies

Mode of learning: Project based

Omdena
Natural Language Processing

Mode of learning: Project based

FreeCodeCamp
Data Analysis with Python

Mode of learning: Online

NPTEL
Data-Base Management System

Mode of learning: Hybrid

NPTEL
Internet of things

Mode of learning: Hybrid